Technical Report — File Conversion System

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1. Introduction

This document demonstrates the highly readable **Markdown → PDF** conversion produced by `tomdclaude`. The goal is not a raw printout of the source but a *report-style document*: cover, automatic table of contents with page numbers, chapter numbering, clean tables and code with syntax highlighting.

The text is justified and hyphenated to improve readability on an A4 page.

1.1 Project goals

1. Convert any format to any other, where that makes sense.
2. Keep high fidelity to Markdown for use with an LLM.
3. Never save a file that fails the integrity checks.

1.2 Constraints

- Everything is **free and local**, with no paid APIs.
- No document uploads.
- Works on a machine without a GPU.

2. Architecture

The system is organized into independent modules. Each converter is a class that inherits from `BaseConverter` and exposes a uniform contract.

2.1 Engine comparison

The following table summarizes the technical options evaluated for the Markdown → PDF path.

Engine	Typographic fidelity	CSS control	Tables	Notes
WeasyPrint	High	Full (CSS)	Great	Chosen for reports
LaTeX/Pandoc	Very high	Indirect	Good	Needs a TeX toolchain
ReportLab	Medium	Manual	Manual	Too low-level

> *WeasyPrint controls the layout with standard CSS, including headers, footers and page numbers in the table of contents.*

2.2 Code example

The heart of the gatekeeper: the conversion is saved only if it is valid.

```
def commit_or_abort(tmp, dst, issues):
    if any(i.blocking for i in issues):
        tmp.unlink(missing_ok=True)   # no file if data was lost
        raise ValueError("conversion rejected by the validator")
    os.replace(tmp, dst)              # atomic commit
```

3. Validation

The *Validator* module compares the source metadata with that of the product.

> **The gatekeeper principle**
> No file is written to the destination until all the integrity checks have passed. Otherwise a detailed error log is produced.

3.1 Implemented checks

- **Heading hierarchy (MD):** the sequence of heading levels must match.
- **PDF integrity:** valid header, at least one page, text in proportion.
- **Rows/columns (XLSX):** no sheet, row or column lost.

4. Conclusions

The Markdown → HTML → PDF path with WeasyPrint produces readable, professional documents, with full control of the style through CSS.